

Insurance Management System — Business & Functional Specification

1. Scope, assumptions, and operating model

This specification describes a generic enterprise insurance operating model suitable for a solo-developer portfolio project that must still feel like a real insurer's internal system. It is written as a business specification, not a technical design.

1.1 Assumptions used throughout

- The system supports multiple insurance lines: auto, health, life, property/home, travel, and business/commercial.
- The system supports three user-facing portals: Customer Portal, Employee Portal, and Agent/Broker Portal.
- The system is role-based and record-based: visibility depends on both role and the current state of the record.
- All workflows are audit-trailed.
- Dates, amounts, and statuses are business-controlled values and must be validated before a record can progress.
- Country-specific regulatory fields are intentionally kept generic. Local legal details should be adapted per jurisdiction.

1.2 Enterprise operating principle

Every record in the system follows the same operational pattern: 1. Create a draft or pending record. 2. Validate required information. 3. Submit to the next business step. 4. Assign ownership to a queue or person. 5. Review, approve, reject, amend, or settle. 6. Store the final decision with audit history. 7. Trigger notifications, documents, and financial effects.

1.3 System-wide statuses

These shared concepts appear across many modules:

- Draft
- Pending Submission
- Submitted
- Under Review
- Approved
- Partially Approved
- Rejected
- Active
- Suspended
- Cancelled

- Lapsed
- Closed
- Paid
- Partially Paid
- Failed
- Archived

1.4 System-wide rule style

For every form, the default expectation is:

- Required fields are clearly marked.
- Inline validation appears immediately for format errors.
- Business-rule validation appears on submit.
- Read-only fields remain visible unless business secrecy requires hiding them.
- Customer-facing screens are simpler and contain fewer operational controls than employee screens.

2. Roles, portals, and visibility model

2.1 Roles

Role	Main purpose	Typical access
Customer	Self-service policyholder or applicant	Own profiles, quotes, claims, payments, documents
Agent	Sells products and manages customers	Customer accounts, quotes, policies, commission views
Broker	Similar to agent but often represents customer interests	Quotes, policies, submissions, commissions
Underwriter	Reviews risk and approves pricing/coverage	Quotes, underwriting decisions, endorsements, referrals
Claims Adjuster	Investigates and settles claims	Claims queues, investigation tools, settlement decisions
Finance Officer	Manages billing, receipts, refunds, collections	Invoices, payments, retries, refunds, ledger impacts
Compliance Officer	Reviews control and regulatory adherence	Audit logs, complaints, legal cases, exception review
Branch Manager	Oversees branch operations and staff	Branch dashboards, approvals, performance, escalations
Admin	Configures masters and users	Roles, users, product setup, document templates

Role	Main purpose	Typical access
Super Admin	Full system authority	All configuration and emergency access

2.2 Portals

Portal	Audience	Content style
Customer Portal	Customers	Simple, guided, task-based
Employee Portal	Internal staff	Dense operational screens with queues and approvals
Agent/Broker Portal	Distribution partners	Sales-oriented screens with customer and quote management

2.3 Visibility rule

An item can be:

- Visible and editable
- Visible and read-only
- Visible but disabled
- Completely hidden

The most important rule is that customer screens must not expose internal underwriting notes, fraud indicators, internal escalation comments, staff assignment data, or settlement reasoning unless intentionally published.

3. Global screen conventions

3.1 Standard page layout

Every major screen should use this pattern:

- Header: page title, key status badge, primary action button
- Summary panel: high-level record identity
- Main tabs or sections: details, timeline, documents, notes, financials, actions
- Side panel or sticky panel: quick facts, ownership, SLA, alerts
- Footer actions: save draft, submit, approve, reject, cancel, close

3.2 Standard table conventions

Column type	Meaning
Identifier	Policy number, claim reference, quote number
Person	Customer, claimant, beneficiary

Column type	Meaning
Dates	Created date, due date, effective date
Amounts	Premium, claim amount, refund amount, outstanding balance
Status	Badge with business meaning
Action	View, edit, approve, reject, assign

3.3 Standard button behavior

Button	Meaning
Save Draft	Keep the record incomplete but stored
Submit	Move the record to the next business step
Approve	Confirm a decision and trigger downstream effects
Reject	Stop the process and record the reason
Cancel	Withdraw the current process or record
Assign	Send a task or record to a person or queue
Add Note	Record internal commentary
Upload Document	Attach supporting evidence
Export	Generate a business report

4. Core data objects

4.1 Customer profile

Contains the identity and contact information of a person or organization, along with household or business relationships and service preferences.

4.2 Quote

A pre-policy pricing proposal that may be accepted, declined, expired, or converted into a policy.

4.3 Policy

The active contractual record that governs coverage, premiums, endorsements, renewals, cancellations, and claims eligibility.

4.4 Claim

A request for benefit or indemnity after a loss event.

4.5 Invoice / premium schedule

A billing record showing what the customer owes and when.

4.6 Payment

Money received or attempted against an invoice, installment, refund, or claim settlement.

4.7 Task

A work item assigned to a user or queue.

4.8 Document

A file attached to a profile, quote, policy, claim, or complaint.

4.9 Notification

A message sent through email, SMS, or in-app alert.

4.10 Audit log

A permanent business history of what changed, who changed it, when, and why.

5. Authentication & authorization

5.1 Why this module exists

Insurance systems handle sensitive personal, financial, and claims data. The system must control access by role, business unit, and record ownership.

5.2 Screens

5.2.1 Sign in

Page title: Sign in

Sections:

- Credentials area
- Help / recovery links

- Security notice

Fields | Field Name | Label | Type | Required | Validation | Who Can Edit | When Read-Only | Notes |
 |---|---|---:|---:|---|---|---|---| | emailOrUsername | Email or Username | text | Yes | Must not be blank |
 User | Never | Accepts either email or account ID | | password | Password | password | Yes | Minimum
 length policy, if configured | User | Never | Masked input | | rememberMe | Remember me | checkbox |
 No | None | User | Never | Optional session persistence |

Buttons | Button Label | Location | Visible To | Enabled When | Action | Confirmation Required | Success
 Outcome | Error Outcome | |---|---|---|---|---|---|---| | Sign In | Primary bottom-right | All users |
 Credentials entered | Authenticates and redirects | No | Lands on role dashboard | Shows invalid
 credential message | | Forgot Password | Under password field | All users | Always | Opens recovery flow
 | No | Starts reset workflow | None |

Show/hide logic | Element | Visible To | Hidden From | Condition | |---|---|---|---| | Password recovery link
 | All | None | Always visible | | MFA prompt | Users with MFA enabled | Users without MFA | After
 credential validation |

Validation

- Email or username cannot be blank.
- Password cannot be blank.
- Account lockout occurs after repeated failed attempts.

Operational example At login, a customer enters their email and password. The system checks the credentials, then prompts for MFA if enabled. After success, the customer lands on their dashboard showing active policies, unpaid invoices, and open claims. If the password is wrong, the user sees a generic authentication error without exposing whether the account exists.

5.3 Authorization model

Each page and action is controlled by:

- Role
- Ownership
- Branch or business unit
- Record state
- Exception permission

5.4 Common mistakes

- Giving internal staff access to all customers instead of only assigned or permitted records.
- Exposing underwriting notes to customers.
- Allowing inactive users to retain portal access.

6. Customer management & profiles

6.1 Purpose

Maintains the master record for an individual or business customer and all linked policies, claims, payments, contacts, and communication preferences.

6.2 Screens

6.2.1 Customer list

Page title: Customers

Sections: search filters, results table, actions toolbar.

Columns | Column | Meaning | Sortable | Filterable | |---|---|---|---| | Customer ID | Internal reference | Yes | Yes | | Full Name / Company Name | Display name | Yes | Yes | | Customer Type | Individual or business | Yes | Yes | | Phone | Primary phone | Yes | Yes | | Email | Primary email | Yes | Yes | | Status | Active / inactive / blocked | Yes | Yes | | Assigned Agent | Responsible sales contact | Yes | Yes | | Open Policies | Count of active policies | Yes | Yes | | Open Claims | Count of active claims | Yes | Yes |

Buttons | Button Label | Location | Visible To | Enabled When | Action | Confirmation Required | Success Outcome | Error Outcome | |---|---|---|---|---|---|---|---| | Create Customer | Top-right | Agent, Broker, Admin, Super Admin | Always | Opens create form | No | New customer draft created | Error if required fields missing | | View | Each row | Allowed roles | Always | Opens profile | No | Shows profile detail | None | | Merge Duplicate | Profile actions | Admin, Super Admin | Duplicate detected | Combines profiles | Yes | Unified customer record | Rollback if merge fails |

6.3 Customer profile sections

- Identity details
- Contact details
- Address history
- Communication preferences
- Household or business relationships
- Policies
- Claims
- Payments
- Documents
- Notes and interaction history

6.4 Full field set

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
customerType	Customer Type	dropdown	Yes	Individual / Business	Agent, Broker, Admin	After creation for compliance-sensitive records	Controls downstream fields
firstName	First Name	text	Conditional	Alphabetic, length limit	Agent, Broker, Admin	After policy issuance if locked	Individual only
lastName	Last Name	text	Conditional	Alphabetic, length limit	Agent, Broker, Admin	After policy issuance if locked	Individual only
companyName	Company Name	text	Conditional	Not blank for business customer	Agent, Broker, Admin	After policy issuance if locked	Business only
dateOfBirth	Date of Birth	date	Conditional	Must be in the past	Agent, Broker, Admin	After underwriting lock	Individual only
nationalId	National ID	text	Conditional	Country format	Agent, Broker, Admin	Usually read-only after verification	Sensitive identity field
phone	Phone	text	Yes	Valid phone format	Agent, Broker, Customer (own profile)	When verified or locked	Primary contact
email	Email	text	Yes	Valid email format	Agent, Broker, Customer (own profile)	When verified or locked	Used for notifications
address	Address	multiline text	Yes	Not blank	Agent, Broker, Customer (own profile)	When policy documents are finalized	Mailing address

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
preferredLanguage	Preferred Language	dropdown	No	Configured list	Customer, Agent	After profile completion if needed	Communication preference
consentToContact	Consent to Contact	checkbox	Yes for marketing	Boolean	Customer, Agent	When legally recorded	Optional by context

6.5 Statuses

- Active
- Inactive
- Blocked for compliance
- Duplicate pending review
- Archived

6.6 Operational example

A broker creates a new customer profile for a family that wants auto and home coverage. The broker enters the family head as the policy contact, adds spouse and dependent records, uploads identity documents, and marks the customer as active. The system flags a possible duplicate because the national ID already exists. The broker cannot complete the record until the duplicate check is reviewed by an admin.

6.7 Common mistakes

- Using one customer record for multiple unrelated people.
- Not separating billing contact from policy holder.
- Forgetting to update addresses after a move.

7. Beneficiaries & dependents

7.1 Purpose

Used mainly for life and health insurance, but can also support any policy that requires related persons.

7.2 Fields

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
relationshipType	Relationship	dropdown	Yes	Spouse, child, parent, other	Customer, Agent, Broker	After policy issue if legal lock applies	Defines dependency
fullName	Full Name	text	Yes	Not blank	Customer, Agent, Broker	After claim submission or policy issue if locked	Beneficiary/dependent name
dateOfBirth	Date of Birth	date	Conditional	Past date	Customer, Agent, Broker	After lock	Required for dependents
allocationPercent	Allocation %	number	Conditional	0–100	Agent, Broker, Customer	After policy issue if required	Beneficiary split
primaryBeneficiary	Primary Beneficiary	checkbox	No	Boolean	Customer, Agent, Broker	After claim or policy lock	One or more may exist depending on product

7.3 Business rule

Allocation totals must equal 100% where product design requires percentage assignment.

7.4 Example

A life customer adds three beneficiaries. The system requires the allocation percentages to total 100%. When the total is 95%, the submit button remains disabled and the screen shows “Beneficiary allocation must total 100%.”

8. Insurance products & plans

8.1 Purpose

Defines what can be sold, under what coverages, at what price logic, and with what exclusions.

8.2 Product structure

A product consists of:

- Product family
- Plan or tier
- Coverage items
- Limits and deductibles
- Eligibility rules
- Underwriting rules
- Required documents
- Pricing factors
- Commission structure

8.3 Product list screen

Column	Meaning	Sortable	Filterable
Product Code	Internal product code	Yes	Yes
Product Name	Display name	Yes	Yes
Line of Business	Auto, health, life, etc.	Yes	Yes
Plan Type	Basic, standard, premium	Yes	Yes
Status	Draft, active, retired	Yes	Yes
Effective From	Start date	Yes	Yes
Effective To	End date	Yes	Yes

8.4 Product fields

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
productCode	Product Code	text	Yes	Unique, no spaces if configured	Admin	After activation	Internal identifier
productName	Product Name	text	Yes	Not blank	Admin	After activation	Marketing name
lineOfBusiness	Line of Business	dropdown	Yes	One of configured lines	Admin	After activation	Controls workflows

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
planName	Plan Name	text	Yes	Not blank	Admin	After activation	Tier name
description	Description	multiline	No	Length limit	Admin	After activation	Sales explanation
status	Status	dropdown	Yes	Draft / Active / Retired	Admin	After approval	Product lifecycle

8.5 Product examples by line

- Auto: vehicle details, driver history, mileage, garaging location
- Health: age, dependent count, hospital network, waiting periods
- Life: age, smoker status, sum assured, beneficiary structure
- Property: occupancy, construction type, value, security features
- Travel: trip dates, destination, trip purpose, baggage cover
- Commercial: business activity, payroll, revenue, asset class, liability limits

9. Quotes & quote comparison

9.1 Purpose

Transforms customer information and risk details into one or more priced offers.

9.2 Quote screen layout

- Customer details
- Risk details
- Coverage selection
- Premium breakdown
- Comparison table
- Underwriting messages
- Submit / save / cancel controls

9.3 Quote comparison table columns

Column	Meaning
Plan	Plan name
Monthly Premium	Recurring payment

Column	Meaning
Annual Premium	Total yearly amount
Deductible	Customer contribution
Coverage Limit	Maximum insurer liability
Key Exclusions	Main exclusions
Recommended	System or agent recommendation flag

9.4 Fields

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
quoteDate	Quote Date	date	Yes	Not future-dated if policy rules require	Agent, Broker, System	After issuance	Quote creation date
desiredStartDate	Desired Start Date	date	Yes	Must be valid based on lead time	Agent, Broker, Customer	After submission	Policy start request
selectedPlan	Plan	dropdown	Yes	Must be eligible	Agent, Broker, Customer	Once submitted	Chosen offer
coverAmount	Coverage Amount	number	Yes	Min/max per product	Agent, Broker, Customer	After underwriting lock	Sum insured
deductible	Deductible	dropdown/number	No	Product-defined values	Agent, Broker	After quote issue	Impacts premium

9.5 Workflow

1. User opens a new quote.
2. Selects customer.
3. Enters risk details.
4. System calculates eligible plans.
5. User compares plans.
6. User submits quote for underwriting or issues it directly if auto-approvable.
7. Quote becomes accepted, expired, declined, or converted.

9.6 Example

An agent quotes auto insurance for a 2021 sedan. The system shows three plans. One plan is hidden because the vehicle age exceeds that plan's limits. The agent compares premium, deductible, and roadside assistance. After the customer selects a plan, the quote is submitted. The system generates a quote number, freezes pricing, and starts underwriting review because the driver has an accident history.

10. Underwriting & risk assessment

10.1 Purpose

Determines whether the insurer accepts the risk, at what price, with what exclusions, and whether special approval is needed.

10.2 Underwriting outcomes

- Accept as quoted
- Accept with modifications
- Refer to senior underwriter
- Reject risk
- Request more information

10.3 Underwriting queue screen

Column	Meaning	Sortable	Filterable
Case Reference	Underwriting case number	Yes	Yes
Customer	Applicant name	Yes	Yes
Product	Line of business	Yes	Yes
Risk Score	Calculated risk indicator	Yes	Yes
Priority	Low / medium / high	Yes	Yes
SLA Due	Review deadline	Yes	Yes
Assigned To	Underwriter owner	Yes	Yes
Status	Open / waiting / decided	Yes	Yes

10.4 Fields

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	No
riskScore	Risk Score	number	System	Calculated	System/ Underwriter override	Usually read-only	Us tria
underwritingDecision	Decision	dropdown	Yes	Accept / refer / reject / modify	Underwriter	After final decision	Co ou
decisionReason	Reason	textarea	Yes if reject or modify	Minimum length if mandatory	Underwriter	After final decision	Au jus
additionalDocumentsRequested	Additional Documents	multi-select	Conditional	Must be from configured list	Underwriter	After request sent	Tri rec cus

10.5 Status transitions

From Status	To Status	Who Can Trigger	Conditions Required	System Actions	Notifications Sent
Pending Review	Under Review	System / Underwriter	Case assigned	Queue timestamps updated	No
Under Review	More Info Required	Underwriter	Missing information	Task created	Customer notification
Under Review	Approved	Underwriter	All criteria met	Quote/policy can proceed	Customer, agent notification
Under Review	Rejected	Underwriter	Risk unacceptable	Record locked	Customer, agent notification
Under Review	Refer to Senior	Underwriter	High-risk threshold reached	Escalation created	Manager notification

10.6 Example

A health applicant discloses a pre-existing condition. The risk score pushes the case into referral. The junior underwriter cannot approve it directly, so they refer it to a senior underwriter. The customer receives a

message requesting medical documents, and the case is placed into a waiting queue until the documents arrive.

11. Policy management & lifecycle

11.1 Purpose

The policy is the active contract and central operational record. Everything else ultimately links to it.

11.2 Policy screens

11.2.1 Policy detail page sections

- Policy summary
- Insured parties
- Coverage and limits
- Premium schedule
- Endorsements
- Documents
- Claims history
- Audit trail
- Notes
- Renewal history

11.3 Policy fields

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
policyNumber	Policy Number	text	Yes	Unique format	System	Never	Generated on activation
policyStartDate	Start Date	date	Yes	Not before acceptance rules	System/ Authorized staff	After activation	Effective date
policyEndDate	End Date	date	Yes	Must be after start date	System/ Authorized staff	After activation	Coverage expiry

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
policyStatus	Status	dropdown	Yes	Draft, quoted, active, suspended, cancelled, lapsed, expired	System	Controlled	Lifecycle state
billingFrequency	Billing Frequency	dropdown	Yes	Monthly, quarterly, annual	Agent, Underwriter, Finance	After activation if locked	Payment schedule

11.4 Policy status transitions

From Status	To Status	Who Can Trigger	Conditions Required	System Actions	Notifications Sent
Quoted	Active	Customer / Agent / System	Accepted quote and payment conditions met	Policy number generated	Customer, agent
Active	Suspended	Finance / Compliance / System	Payment failure or compliance issue	Coverage temporarily stopped	Customer, internal team
Active	Cancelled	Customer / Company	Cancellation rules satisfied	Premium recalculated	Customer, finance
Active	Lapsed	System	Renewal not completed by deadline	Coverage ends	Customer
Suspended	Reinstated	Finance / Authorized staff	Outstanding items resolved	Coverage resumes	Customer

11.5 Important business rules

- A policy cannot be activated without meeting minimum underwriting and payment requirements.
- Endorsements must preserve historical values.
- Cancellation effective dates must respect product and legal notice periods.
- A lapsed policy may require a new underwriting review to reinstate.

11.6 Example

A customer pays the first premium installment for auto insurance. The policy activates automatically. The policy detail page then shows “Active” in a green badge, the coverage tab becomes fully visible, and the

renewal countdown appears. Three months later, if payment fails repeatedly, the policy moves to suspended or lapsed depending on the collection rule set.

12. Policy endorsements & amendments

12.1 Purpose

Changes an active policy without replacing it.

12.2 Common endorsement types

- Address change
- Vehicle replacement
- Beneficiary update
- Coverage increase or reduction
- Named insured change
- Installment plan change
- Add/remove dependent

12.3 Endorsement workflow

1. Initiated by customer, agent, or authorized staff.
2. User selects endorsement type.
3. System shows only relevant fields.
4. Validation checks prior policy constraints.
5. Premium impact is calculated.
6. Endorsement is approved, rejected, or pending review.
7. Policy history keeps the old and new values.

12.4 Example

A customer changes their home address. The endorsement form only shows address-related fields. The system recalculates the risk because the new area has different flood exposure. The endorsement becomes pending underwriting review before the policy can be updated.

13. Claims module

13.1 Purpose

Handles loss reporting, triage, investigation, approval, partial approval, rejection, and settlement.

13.2 Claims submission screen

Sections

- Claimant identity
- Related policy selection
- Incident details
- Damage / loss details
- Supporting documents
- Estimated amount
- Declaration and submission

Fields

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
policyNumber	Policy Number	dropdown	Yes	Must be active or eligible	Customer, Agent, Claims staff	After submission	Must belong to claimant
incidentDate	Incident Date	date	Yes	Cannot be in the future	Customer, Agent	After submission	Core validity check
incidentTime	Incident Time	time	No	Valid time format	Customer, Agent	After submission	Optional
incidentLocation	Incident Location	text	Yes	Not blank	Customer, Agent	After submission	Loss site
incidentDescription	Description of Loss	textarea	Yes	Minimum length, maximum length	Customer, Agent	After submission	Narrative
estimatedLossAmount	Estimated Loss Amount	number	No	Non-negative	Customer, Agent	After investigation begins	Initial estimate
documents	Supporting Documents	file upload	Yes in most products	Allowed types and size limits	Customer, Agent	After final submission	Photos, reports, proof

13.3 Claims statuses

- Draft
- Submitted
- Under Review

- More Information Required
- Approved
- Partially Approved
- Rejected
- Payment Pending
- Paid
- Closed
- Withdrawn

13.4 Claim status transitions

From Status	To Status	Who Can Trigger	Conditions Required	System Actions	Notifications Sent
Draft	Submitted	Customer / Agent	Required fields complete	Claim number generated	Customer, adjuster queue
Submitted	Under Review	System / Adjuster	Assignment completed	SLA timer starts	Internal notification
Under Review	More Information Required	Adjuster	Missing evidence	Task created	Customer notification
Under Review	Approved	Adjuster / Manager	Coverage and liability confirmed	Payment request created	Customer, finance
Under Review	Partially Approved	Adjuster / Manager	Only part of claim is covered	Settlement amount updated	Customer
Under Review	Rejected	Adjuster / Manager	Not covered or invalid	Claim locked	Customer

13.5 Buttons

Button Label	Location	Visible To	Enabled When	Action	Confirmation Required	Success Outcome	Error Outcome
Submit Claim	Bottom of form	Customer, Agent	All mandatory fields valid	Submits claim	Yes if product requires declaration	Claim submitted	Field validation errors
Request More Info	Claim review panel	Adjuster	Review open	Sends info request	Yes	Claim paused	None

Button Label	Location	Visible To	Enabled When	Action	Confirmation Required	Success Outcome	Error Outcome
Approve	Claim decision panel	Adjuster, Manager	Decision criteria satisfied	Approves claim	Yes	Settlement workflow begins	Permission/validation failure
Reject	Claim decision panel	Adjuster, Manager	Review open	Rejects claim	Yes	Claim closed/rejected	None
Pay Claim	Finance panel	Finance Officer	Approved and payment ready	Initiates settlement	Yes	Payment created	Payment failure

13.6 Example

A customer submits a motor claim after a collision. The system blocks submission until the incident date is not future-dated, at least one photo is attached, and the declaration checkbox is ticked. After submission, the claim enters the adjuster queue with priority based on claim amount. The adjuster requests a police report, the customer uploads it, and later the claim is approved for partial payment because wear-and-tear items are excluded.

14. Claims assignment, investigation, approval, settlement, and fraud scoring

14.1 Assignment rule

Claims are assigned by line of business, severity, geography, workload, and SLA priority.

14.2 Investigation sections

- Claim facts
- Coverage check
- Liability analysis
- Damage assessment
- Witness / document review
- Fraud indicators
- Decision notes

14.3 Fraud scoring

A fraud score is not a final decision. It is a triage indicator.

Common triggers:

- Repeated claims in short periods
- Mismatch between incident timing and reporting time
- High-value claim with weak evidence
- Inconsistent narratives
- Similar claims across related customers

14.4 Settlement outcomes

- Full settlement
- Partial settlement
- Payment by repairer / provider
- Rejection
- Recovery / subrogation referral

14.5 Example

A claim is flagged because it was filed 45 minutes after the policy started and the reported damage amount is unusually high. The fraud score causes escalation to a senior adjuster. Investigation reveals a valid loss but part of the invoice is unrelated to the insured event, so the claim is approved partially.

15. Billing, premium calculation, installments, failures, refunds, collections, accounting**15.1 Purpose**

Ensures the insurer collects the correct premium amount on time and records financial movement accurately.

15.2 Billing screen sections

- Policy billing summary
- Installment schedule
- Open invoices
- Payments received
- Failed payments
- Refunds
- Collection actions
- Ledger impact

15.3 Premium calculation factors

- Product base rate
- Risk score
- Coverage amount

- Deductible
- Installment frequency
- Discounts and surcharges
- Taxes and fees

15.4 Invoice fields

Field Name	Label	Type	Required	Validation	Who Can Edit	When Read-Only	Notes
invoiceNumber	Invoice Number	text	Yes	Unique generated value	System	Never	Billing reference
dueDate	Due Date	date	Yes	Must follow billing schedule	System/ Finance	After issue	Payment deadline
amountDue	Amount Due	number	Yes	Non-negative	System/ Finance	After issue	Includes taxes/ fees if applicable
invoiceStatus	Status	dropdown	Yes	Open / paid / failed / cancelled / reversed	System	Controlled	Billing lifecycle

15.5 Payment failure and retry logic

A failed payment can trigger:

- Retry attempt
- Customer reminder
- Manual collections task
- Policy suspension if threshold reached
- Finance escalation

15.6 Example

A monthly premium fails because the card is expired. The system creates a failed payment record, sends the customer an email, schedules a retry after 48 hours, and creates a collections task if the second attempt also fails. If the grace period expires, the policy can move to suspended or lapsed depending on product rules.

16. Renewals, cancellations, suspensions, reinstatements, lapses

16.1 Renewals

- Start with a renewal notice.
- Generate updated premium.
- Collect customer decision.
- Underwrite again only if risk changed or rules require.

16.2 Cancellations

Customer-initiated: the customer requests to end coverage.

Company-initiated: fraud, non-payment, regulatory non-compliance, or product withdrawal.

16.3 Suspensions

Temporary stop in coverage, usually after non-payment or compliance issue.

16.4 Reinstatements

Coverage resumes only after outstanding conditions are resolved.

16.5 Lapses

Policy ends because renewal or payment requirements were not met.

16.6 Example

A renewal notice is sent 30 days before expiry. The customer does not pay. A reminder is sent. The policy lapses after the grace period. Later, a reinstatement request is made, but the system requires fresh underwriting because too much time has passed.

17. Agents, brokers, commissions, branch management

17.1 Purpose

Supports the distribution network and compensation tracking.

17.2 Agent/Broker screens

- Profile
- Assigned customers
- Quotes created
- Policies sold
- Commissions earned

- Performance metrics

17.3 Commission rules

Commissions may vary by:

- Product line
- Policy duration
- New business vs renewal
- Cancellation clawback
- Premium collected vs premium written

17.4 Example

An agent receives commission only after premium is collected, not merely after policy issuance. If the policy is cancelled early, a clawback entry is created for unearned commission.

18. Support, documents, notes, complaints, disputes

18.1 Purpose

Tracks service interactions, evidence, internal discussion, and formal complaints.

18.2 Documents

- Identity documents
- Policy schedules
- Payment receipts
- Claim evidence
- Medical or repair reports
- Complaint letters

18.3 Notes

Internal notes are staff-only. Customer-visible notes must be deliberately published.

18.4 Complaints flow

1. Complaint logged.
2. Assigned to team.
3. Investigation performed.
4. Response drafted.
5. Complaint resolved or escalated.
6. Audit trail retained.

18.5 Example

A customer complains that a claim was settled too slowly. The complaint is logged separately from the claim, assigned an SLA due date, and resolved with a formal response letter approved by a manager.

19. Governance, approvals, audit, compliance, legal, SLA, tasks

19.1 Approval pipeline

Used when an action exceeds ordinary authority.

19.2 Audit log content

Field	Meaning
Action	What changed
Actor	Who changed it
Timestamp	When it happened
Before / After	Old and new values
Reason	Why the action happened

19.3 SLA management

Every case can have a due date, escalation path, and breach status.

19.4 Example

A high-value claim requires manager approval. If the manager does not act within SLA, the system escalates to the branch manager and marks the case as overdue.

20. Analytics, dashboards, and KPIs

20.1 Per-role dashboards

- Customer: policies, claims, payments, alerts
- Agent: sales pipeline, commissions, customer tasks
- Underwriter: referral queue, risk distribution, SLA compliance
- Claims: open claims, age of claims, settlement speed
- Finance: receivables, failures, aging, refunds
- Management: portfolio growth, loss ratio, cancellation rate

20.2 KPI examples

- Quote-to-policy conversion rate
 - Average claim turnaround time
 - Payment failure rate
 - Renewal retention rate
 - Complaint resolution SLA compliance
 - Fraud detection referral rate
-

21. Insurance line-specific differences

21.1 Auto insurance

Unique fields: vehicle make, model, year, VIN, plate number, driver history, garaging address, annual mileage.

Unique risks: driver age, accident history, vehicle use, theft exposure.

Common claims: collision, theft, third-party damage, windshield damage.

Required documents: driver ID, vehicle registration, photos, police report for major losses.

21.2 Health insurance

Unique fields: dependent count, plan network, waiting periods, pre-existing conditions, hospital preference.

Unique risks: age, family size, chronic conditions, policy selection timing.

Claims: outpatient, inpatient, pharmacy, maternity, emergency.

Required documents: medical invoices, discharge summary, doctor report, referral where needed.

21.3 Life insurance

Unique fields: smoker status, occupation risk, sum assured, beneficiaries, medical questionnaire, medical exam requirement.

Unique risks: age, health history, occupation, lifestyle, coverage amount.

Claims: death benefit, maturity benefit, disability riders if included.

Required documents: death certificate, claimant identity, beneficiary proof, policy document.

21.4 Property / home insurance

Unique fields: property type, construction material, occupancy, security devices, rebuild value, contents value.

Unique risks: fire, theft, flood, earthquake, vacant property.

Claims: structural damage, contents loss, temporary accommodation.

Required documents: photos, repair estimates, police/fire reports where relevant.

21.5 Travel insurance

Unique fields: trip dates, destination, purpose of travel, traveler age, trip cost, baggage value.

Unique risks: destination risk, trip cancellation, medical emergency abroad.

Claims: medical expenses, baggage loss, delayed flight, cancellation.

Required documents: booking proof, boarding pass, receipts, airline report.

21.6 Business / commercial insurance

Unique fields: business type, revenue, payroll, assets, number of employees, liability limits, location exposure.

Unique risks: operational hazard, professional liability, property exposure, employee injury.

Claims: property loss, liability claim, business interruption, equipment damage.

Required documents: financial records, incident reports, contracts, police reports, repair invoices.

22. MVP vs standard vs advanced scope

22.1 MVP for a solo developer portfolio

- Authentication and role-based portals
- Customer management
- Products, quotes, policies, claims, billing
- Basic underwriting and approval flows
- Notifications and document uploads
- Audit log and dashboards

22.2 Standard enterprise scope

- Commission management
- SLA tracking
- Complaints and disputes
- Collections and retries
- Partial approvals and endorsements
- Branch management

22.3 Advanced enterprise scope

- Fraud scoring
- Reinsurance support
- Legal case management
- Historical tracking and archive rules
- Sophisticated workflow routing
- Multi-entity accounting integration

23. Common implementation mistakes to avoid in the business design

- Mixing customer-facing and employee-facing fields on the same screen without access rules.
- Treating status as a visual label instead of a workflow gate.
- Allowing edits after a policy or claim is locked.
- Failing to preserve historical values after endorsements.
- Sending operational messages without recording an audit trail.
- Letting claims settle without referencing coverage and deductible logic.

24. Final design principle

A real insurance system is not just a set of forms. It is a controlled operational machine where every action has a business consequence: a status change, an assignment, a notification, a document, a financial impact, and an audit trail.

This specification should be used as the operational blueprint for implementation, screen design, workflow design, and domain modeling.